Mechano-Sterol Homeostasis, Ciliary Hedgehog-YAP/ERK Signaling, and Sequential Post-Transcriptional Clocks in Human Neocortical Expansion: A Multi-Omic Dynamical Systems Framework

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Abstract

The evolutionary expansion and folding of the human neocortex depend on how delaminated HOPX$^+$ outer radial glia (oRGs) keep dividing inside the outer subventricular zone (oSVZ) during mid-gestation (gestational weeks 13–22, GW13–22), and how they step down on time into CRYAB$^+$ truncated radial glia (tRGs) and gliogenic tri-potential intermediate progenitor cells (Tri-IPCs). When we look closely at this timeline, two hard questions show up right away. First, how do delaminated human oRGs keep high nuclear YAP (86% vs. 4% in smooth-brained mice) and selective mTORC1 (pS6) activity inside a very soft tissue niche ($E < 1,000$ Pa) after losing their apical adherens junctions? Second, how does this growth engine turn itself off in late gestation without trapping migrating neurons in the subplate or causing runaway tumor-like growth?

In this paper, I bring together published patch-clamp electrophysiology, atomic force microscopy (AFM), paleogenomic, primary human fetal single-cell transcriptomic (NCBI GEO: GSE162170, $N = 12,634$ radial glia), and cross-linking immunoprecipitation (ENCODE primary eCLIP narrowPeaks and neural iCLIP) measurements to build and test a seven-module (10-state) coupled ordinary differential equation (ODE) framework. Building on the experimental studies that mapped these individual pathways, I connect four areas that are usually studied on their own: (1) actomyosin-driven PIEZO1 mechanotransduction and upstream *de novo* sterol synthesis inside a two-sided membrane cholesterol raft window; (2) the *Homo sapiens*-enriched *pancMEIS1*/MEIS1–KLF6–SCAP/SEC24D sterol-ciliogenesis amplifier; (3) the push-and-pull kinase-morphogen switch between neurogenic ERK/PKA signaling and gliogenic Sonic Hedgehog (SHH)–YAP/TAZ signaling; and (4) a two-stage post-transcriptional RNA-binding protein (RBP) relay moving from first-trimester IGF2BP1 silencing to mid- and late-gestation IGF2BP2/3–SYNCRIP competition.

Testing this system across calibration anchors, module ablations, perturbation benchmarks, and primary human multi-omic datasets shows how the pieces fit together. In the ODE benchmarks, chemical PIEZO1 activation (Yoda1) cannot bypass a downstream KLF6/SCAP sterol block, adding exogenous cholesterol-MβCD (10 μg/ml) rescues *Piezo1* loss-of-function (1.8× neuronal differentiation), excess cholesterol loading (0.15–1.0 mM) shifts the tRG switch earlier (from GW22.1 to GW19.3) by dampening PIEZO1 while saturating ciliary SHH, forcing mid-gestation IGF2BP1 expression recreates subplate migration arrest (1.011 cells/hr), and active SYNCRIP mRNP licensing is needed to shut down ERK/PKA and trigger the GW22–26 tRG switch. In primary human fetal oRGs (GSE162170), neighborhood metacell pooling ($n = 80$) confirms positive covariance between *PIEZO1*/*YAP1* and the Paleo-Sterol Module (*PIEZO1*: Spearman $\rho = 0.4935, p = 3.29 \times 10^{-6}$; *YAP1*: Pearson

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$r = 0.5029, p = 1.99 \times 10^{-6}$). The model's Termination Score ($T_{\text{term}} = 0.55Y + 0.45H + 0.25S$) tracks tRG/gliogenic induction (*CRYAB*, *EGFR*, *SOX9*: $r = 0.5097, p = 1.37 \times 10^{-6}$) and neurogenic *HOPX* decline ($r = -0.2840, p = 0.0107$; single-cell $N = 12,634, p < 10^{-150}$). Across GW16–GW24, endogenous *IGF2BP1* drops 58.8-fold while *CRYAB* surges 23.8-fold, and ENCODE eCLIP narrowPeaks confirm direct RBP binding on *MYC*, *MAP2K1*, *DHCR7*, *MEIS1*, *KLF6*, *YAP1*, *SOX9*, and *EGFR* with zero peaks on negative controls (*PIEZO1*, *SCAP*, *STOML3*). Across an $N = 300$ Latin Hypercube sensitivity sweep, 92.0% of runs switch inside the normal GW17–26 window (mean 22.4 ± 3.5 GW), and phase-plane analysis shows that niche stiffening and SYNCRIP licensing drive a saddle-node bifurcation that closes the neurogenic attractor.

Keywords: Neocortical expansion, outer radial glia, PIEZO1, cholesterol homeostasis, KLF6, SCAP, MEIS1, IGF2BP1/2/3, SYNCRIP, ERK/PKA signaling, YAP/TAZ, Sonic Hedgehog, dynamical systems modeling, single-cell multi-omics.

1 Introduction

Over the past decade, developmental neurobiology has mapped out hundreds of genes that control how neural progenitors divide, how newborn neurons migrate, and how cortical layers form. Even with all of this experimental progress, these pathways are still mostly studied in separate tracks. Evolutionary transcriptomics, cell mechanics, cholesterol metabolism, kinase signaling, and RNA-binding proteins are rarely put together into one working mathematical system. When we set out to build a unified model of how the human cortex grows, we need to look at where these pathways meet and make sure every parameter matches published measurements for the right cell type, developmental stage, and causal direction.

During human cortical development, ventricular radial glia (vRGs) lower their apical adherens junction proteins between GW11 and GW14 and move out into the basal outer subventricular zone (oSVZ). There, they become HOPX^+ outer radial glia (oRGs, or basal radial glia, bRGs), keeping only a long basal fiber attached to the pial basement membrane through integrin $\alpha_6\beta_1$ complexes and PALMD-dependent protrusions [1, 2].

Where Simple Linear Models Run Into Trouble. When vRGs delaminate to become oRGs, two major puzzles show up that are worth looking into carefully:

1. **The Soft-Niche Proliferation Paradox:** In standard epithelial cells, losing apical junctions while sitting in a very soft extracellular matrix ($E < 1,000$ Pa) turns on the Hippo kinases MST1/2 and LATS1/2. That traps YAP/TAZ in the cytoplasm and stops cell division. Yet in the human fetal cortex at GW13–14, 86% of SOX2^+ basal progenitors keep active YAP inside the nucleus (compared to just 4% in smooth-brained mouse basal progenitors) [3]. At the same time, they show selective mTORC1 (phospho-S6, pS6) activity that is missing in vRGs, TBR2^+ intermediate progenitor cells (IPCs), and CRYAB^+ truncated radial glia (tRGs) [4].

2. **The Non-Oncogenic Termination Paradox:** Human oRGs need steady nuclear YAP, ERK/PKA, and mTORC1 activity to run multiple rounds of self-renewal. However, if YAP/TAZ stays high forever [5] or if early RNA-binding proteins like IGF2BP1 (IMP1) are kept on during mid-neurogenesis [6], development runs into trouble. Newborn neurons either get held up in the subplate at a linear rate (1.011 cells/hr) or progenitors step down early into ependymal and glial fates [7].

What This Paper Addresses. Because the experimental papers covering these topics through 2026 were designed to study individual mechanisms in different model systems, combining them into one mathematical model requires updating five natural simplifying assumptions that can arise when data from neighboring cell types or stages are blended together (Table 1). In this work, I set up a 7-module (10-state) dynamical system that resolves these cross-system

differences, separates calibration anchors from emergent predictions, tests the circuit directly against primary human fetal scRNA-seq ($N = 12,634$ radial glia) and ENCODE eCLIP binding maps, and defines the exact theoretical boundaries and wet-lab protocols for Phase-II benchtop verification.

Table 1: Cross-Disciplinary Parameter Provenance and Updates to Common Simplifying Assumptions in Cortical Systems Modeling.

Domain	Common Simplifying Assumption	Published Primary Measurement	How Our Model Handles It
PIEZO1 Electrophysiology	Using glia-lineage patch values ($I_{\max} = 503.8 \pm 183$ pA, $\tau_{\text{inact}} = 37.3 \pm 2.75$ ms, 0.4 μm steps, 13/14 cells) [8] for radial glia.	Those numbers come from oligodendrocyte progenitor cells (OPCs) [8]. Human NSPCs (SC23/SC27) show 0.25 μm indentation steps (-80 mV) and cell-attached suction currents in 91/107 cells (85.05%) with $P_{50} = -13.4$ mmHg, $s = 4.75$ mmHg, and $E_{\text{rev}} = -0.75 \pm 0.67$ mV [9].	Uses human NSPC Boltzmann suction kinetics paired with internal Myosin-II traction Ca^{2+} flickers [10].
Cortical Niche Biomechanics	Using E12.5 mouse pial surface AFM (52.46 Pa lateral vs. 128.1 Pa dorsal) [11] as a proxy for the deep human oSVZ.	Pial surface numbers measure the marginal zone for Cajal-Retzius migration [11]. Direct slice AFM of living mouse VZ/SVZ stiffens from 46.8–74.0 Pa (E12.5) to 149.3–169.3 Pa (E18.5) with an E16.5 IZ peak of 216.7 Pa [12], while glyoxal-fixed ferret zones span 423 to 943 Pa [13].	Uses a zone-specific stiffening curve across the VZ, SVZ, and IZ that gradually raises mechanical load on delaminated oRGs.
PIEZO1–Sterol Causality	Placing KLF6/SCAP sterol synthesis upstream to pack the membrane and boost PIEZO1, which predicts Yoda1 rescues KLF6/SCAP KO.	PIEZO1 sits upstream of *de novo* sterol genes (*Hmgcr*, *Dhcr7*); adding cholesterol-MβCD (10 μg/ml) rescues *Piezo1* KO differentiation (1.8× neuronal, 1.25× astrocytic) [14]. Both MβCD depletion ($IC_{50} = 1.12$ mM) and excess cholesterol ($IC_{50} = 0.15$ mM) reduce PIEZO1 activity [15, 16].	Models membrane sterol gating as a two-sided window $\Phi_{\text{raft}}(C)$, placing KLF6/*pancMEIS1*/SCAP sterol flux downstream to feed primary cilia, SHH/GLI3, and mTORC1 [17, 18].
Two-Stage RBP Temporal Clock	Keeping IGF2BP1 active in mid-gestation (GW16–22) oRGs to directly stabilize *PIEZO1* and *SCAP* mRNAs.	Endogenous IGF2BP1 turns off before mid-gestation ($<$ GW12) [19, 20]. Keeping it on at mid-stage (T2Imp1) traps cells in the subplate at 1.011 cells/hr [6]. CLIP-seq maps RBP competition to *MYC*, *GLI3*, and metabolic TFs rather than ion channels [21].	Builds a Two-Stage Relay: Stage I IGF2BP1 shutoff allows delamination; Stage II IGF2BP2/3 vs. SYNCRIP competition controls *MYC*, *GLI3*, and pro-ERK transcripts.
Kinase & Morphogen Fate Logic	Treating high YAP/TAZ as a block on tRG formation while falling ERK/PKA passively ends expansion.	ERK and PKA keep radial glia in a neurogenic state (N-RGs) and hold back YAP via LATS1/2, while rising ciliary SHH, nuclear YAP/TAZ, and SYNCRIP actively shut down ERK/PKA at E16.5/GW22–26 to form $CRYAB^{+}$ tRGs and gliogenic Tri-IPCs [7].	Connects sterol-driven ciliary SHH, matrix-driven YAP/TAZ, and SYNCRIP licensing ($\omega_S S$) to active shutoff of neurogenic kinase activity K_{EP}.

2 Biophysical, Metabolic, and Post-Transcriptional Parameterization

2.1 Module 1: Human NSPC PIEZO1 Kinetics and Germinal Niche Mechanics

Let us first look at the published channel measurements in human neural stem/progenitor cells (hNSPC lines SC23 and SC27). Whole-cell patch-clamp recordings during piezoelectric poking (0.25 μm steps at -80 mV) and cell-attached suction steps (0 to -30 mmHg in 3-mmHg steps) trigger stretch-activated currents in 85.05% of patches (91 out of 107 cells) [9]. These currents follow a clean Boltzmann suction curve with a half-activation pressure of $P_{50} = -13.4$ mmHg, a slope factor of $s = 4.75$ mmHg, and a reversal potential of $E_{\text{rev}} = -0.75 \pm 0.67$ mV, and they can be reversibly blocked by the tarantula venom peptide GsMTx-4 (5–100 μM). Even when there is no outside fluid flow, the cell's own Myosin-II pulling forces on focal adhesions ($\sigma_{\text{myo}} \sim 1.8$–$2.5$ kPa) are enough to trigger local PIEZO1 Ca^{2+} flickers [10].

On the tissue side, direct AFM poking (3 μm depth) of living, agar-framed coronal slices across mouse development (E12.5 to E18.5) shows that the deep germinal zones steadily stiffen over time [12]. The dorsal VZ Young's modulus (E) climbs from 46.8 ± 10.4 Pa (E12.5) to 72.3 ± 10.7 Pa (E14.5), 87.8 ± 12.3 Pa (E16.5), and 149.3 ± 15.0 Pa (E18.5). At the same time, SVZ stiffness rises from 74.0 ± 11.5 Pa (E12.5) to 133.3 ± 23.3 Pa (E16.5) and 169.3 ± 18.7 Pa (E18.5), with an intermediate zone (IZ) peak of 216.7 ± 23.4 Pa at E16.5. Also notice what happens when single cells are tested on their own: dissociated $Pax6^+$ apical radial glia actually soften from 149.2 ± 27.3 Pa (E14.5) down to 68.15 ± 20.6 Pa (E18.5), whereas newborn $Tuj1^+$ neurons stiffen from 154.7 ± 11.8 Pa up to 230.5 ± 20.0 Pa as their microtubules build up. That tells us the overall tissue stiffening comes from extracellular matrix (ECM) remodeling and tighter cell packing [12]. In the folded ferret cortex (E35 and P0 slices fixed in 3% glyoxal, which raises raw numbers while keeping the relative layer pattern intact), deep VZ/iSVZ zones measure 423 ± 19 Pa and 538 ± 49 Pa, climbing to 768 ± 51 Pa, 743 ± 85 Pa, and 943 ± 136 Pa in the outer zones [13].

2.2 Module 2: Two-Sided Sterol Gating and the Human KLF6–MEIS1–SCAP Amplifier

Knocking out *Piezo1* in E10.5 mouse neural stem cells drops *Hmgcr* and *de novo* cholesterol synthesis genes—along with sterol-modifying enzymes (*Cyp46a1*, *Ch25h*, *Soat2*)—by more than 1.5-fold, draining free cholesterol and lipid droplets inside the cell [14]. Giving *Piezo1*-null NSCs 10 μg/ml of exogenous cholesterol bound to methyl-β-cyclodextrin (cholesterol-MβCD) rescues their differentiation (1.8× neuronal and 1.25× astrocytic increase). This shows that cholesterol synthesis sits downstream of PIEZO1 Ca^{2+} entry. At the same time, coarse-grained molecular dynamics (CG-MD) and patch-clamp tests show that membrane cholesterol controls PIEZO1 through a two-sided window: stripping cholesterol away with extracellular MβCD ($IC_{50} = 1.12$ mM) breaks apart STOML3 lipid rafts and scatters PIEZO1 clusters [15, 16], while packing in too much exogenous cholesterol ($IC_{50} = 0.15$ mM, or 100 μg/ml) stiffens the lipid bilayer, flattens the channel's nanodome curve, and dampens both Yoda1-driven Ca^{2+} entry and stretch currents [15].

In human $HOPX^+$ oRGs, this upstream PIEZO1 sterol signal gets boosted by an evolutionarily selected gene network [17, 18]. KLF6 sits on a radial-glia super-enhancer promoter that drives *PTPRZ1*, *RB1CC1*, and *BHLHE40* in genomic regions free of archaic introgression, while a modern human variant (`chr10:3936512-G-C`, hg38) strengthens GLI3 binding at the *KLF6* locus [17]. This loop works alongside a fixed *Homo sapiens* amino acid change in the sterol sensor SCAP, positive selection on the COPII coat protein SEC24D, shifted SREBF2 binding near *PALMD*, and human-biased R-loop activation of *MEIS1* by its promoter-associated noncoding

RNA (*pancMEIS1*), which directly binds the *DHCR7* promoter to coordinate cholesterol synthesis and primary cilia growth [18]. Rather than piling up in the outer plasma membrane where it would dampen PIEZO1, this extra cholesterol is shipped out through SCAP/SEC24D vesicles to build the primary cilium (helping Smoothened sterol modification and GLI3 processing), grow $PALMD^+$ basal fibers, and keep oRG-specific mTORC1 (pS6) running.

2.3 Module 3: The Two-Stage RBP Relay and the ERK/PKA–SHH/YAP Switch

Endogenous IGF2BP1 (IMP1) is only expressed in early first-trimester neuroepithelial cells (< GW12 in humans; E10.5–E12.5 in mice) and is turned off before mid-gestation oRG expansion begins [19,20]. When *Imp1* is forced to stay on during mid-neurogenesis using the TEMPO lineage-tracing tool (T2Imp1, electroporated at E12.5 and imaged at E14.5 for 48 h), newborn cells get trapped in the IZ–SVZ–VZ and subplate zone (SPZ) at steady linear rates of 1.011 cells/hr (RFP^+) and 0.747 cells/hr (CFP^+, $r > 0.89$, $p < 0.05$) [6]. Put simply, Stage I shutoff of IGF2BP1 opens the gate so oRGs can delaminate and neurons can migrate cleanly. During Stage II (GW13–26), its sister proteins IGF2BP2 and IGF2BP3 stay on to protect m^6A-marked *MYC*, *GLI3*, and metabolic mRNAs in oRGs until newborn neurons pile up and release GDF11 chalone feedback [22]. That GDF11 signal raises SYNCRIP (hnRNPR) inside the shared SYNCRIP–HNRNPU–YBX1–DHX9 mRNP complex [21,23].

Meanwhile, whether a radial glial cell stays neurogenic or steps down into glia comes down to a tug-of-war between neurogenic ERK/PKA signaling and gliogenic SHH–YAP/TAZ signaling [7]. During mid-gestation, PTPRZ1/FGFR-driven ERK1/2 and cAMP-PKA (which phosphorylates LATS1/2 to keep nuclear YAP from getting too high) keep radial glia in their neurogenic state (N-RGs). By late gestation (E16.5 in mice; GW22–26 in humans), stiffer tissue, sterol-fed ciliary SHH signaling, and SYNCRIP-driven breakdown of pro-ERK mRNAs push the combined YAP/TAZ, SHH, and SYNCRIP score past a tipping point (Θ_{switch}). That shuts down ERK/PKA and pS6/mTORC1 and guides the cells into becoming $CRYAB^+$ tRGs and gliogenic Tri-IPCs [7]. Table 2 puts all of these published measurements in one place.

Table 2: Summary of Published Experimental Measurements Used Across Modules.

Category	Metric	Measured Empirical Range	Experimental Setup
Mouse Germinal Mechanics	VZ, SVZ, & IZ Young's Modulus (E)	VZ: 46.8 ± 10.4 Pa (E12.5) $\rightarrow 149.3 \pm 15.0$ Pa (E18.5); SVZ: 74.0 ± 11.5 Pa $\rightarrow 169.3 \pm 18.7$ Pa; IZ Peak: 216.7 ± 23.4 Pa (E16.5)	Living agar-framed coronal slices, 3 μm AFM indentation [12].
Single-Cell Soma Mechanics	Progenitor vs. Neuronal Soma Stiffness (E)	$Pax6^+$ aRG: 149.2 ± 27.3 Pa (E14.5) $\rightarrow 68.15 \pm 20.6$ Pa (E18.5); $Tuj1^+$ Neuron: 154.7 ± 11.8 Pa $\rightarrow 230.5 \pm 20.0$ Pa	Dissociated mouse cortical cells with post-AFM staining [12].
Ferret Cortical Mechanics	Apico-Basal Cortical Gradient (E)	Deep VZ/iSVZ: 423 ± 19 Pa to 538 ± 49 Pa; Outer oSVZ/IZ/CP: 768 ± 51 Pa to 943 ± 136 Pa	3% glyoxal-fixed coronal vibratome sections [13].
hNSPC PIEZO1 Biophysics	Indentation & Suction Boltzmann Kinetics	0.25 μm steps (-80 mV); 85.05% responsive (91/107 cells); $P_{50} = -13.4$ mmHg, $s = 4.75$ mmHg, $E_{\text{rev}} = -0.75 \pm 0.67$ mV	Human fetal neural stem/progenitor lines SC23 and SC27 [9].
Two-Sided Sterol Gating	Depletion vs. Excess Inhibition IC_{50}	MβCD Depletion: $IC_{50} = 1.12$ mM; Exogenous Cholesterol Inhibition: $IC_{50} = 0.15$ mM (100 μg/ml)	Human PIEZO1-expressing cells, patch-clamp, and CG-MD [15, 16].
PIEZO1 Sterol & Fate Output	Astrogenesis Suppression & Sterol Rescue	PIEZO1 KD raises $GFAP^+$ cells from $9.8 \pm 0.5\%$ to $14.2 \pm 0.7\%$ (+44.9%); 10 μg/ml Chol-MβCD rescues KO (1.8× neuronal)	SC23/SC27 hNSPCs and E10.5 *Piezo1* KO mouse NSCs [9, 14].
Imp1 TEMPO Kinetics	Subplate Trapping Slopes	T2Imp1 RFP^+ slope = 1.011 cells/hr; CFP^+ slope = 0.747 cells/hr ($r > 0.89$, $p < 0.05$)	48-h slice imaging (30-min steps) after E12.5 IUE [6].
Signaling & Lineage Switch	Nuclear YAP, pS6, & ERK/PKA–SHH/YAP	Nuclear YAP: 86% human vs. 4% mouse bRGs; selective pS6 in $HOPX^+$ oRGs; GW22–26 SHH+YAP+SYNCRIP switch to $CRYAB^+$ tRG	Human GW13–26 fetal cortex and *Map2k1/2*-dcko / $Smo^{F/F}$ atlases [3, 4, 7].

3 Mathematical Formulation of the 7-Module (10-State) Coupled ODE System

We can now write out the cell and population dynamics of human radial glia across gestational time $t \in [\text{GW10}, \text{GW26}]$ (where 1 GW = 168 hours is the base time unit) as seven coupled ODE modules tracking 10 state variables. All intracellular signaling states are normalized relative to their mid-gestation physiological baseline: raft-available intracellular cholesterol $C(t)$ (normalized to baseline $C_0 = 1.0$), nuclear YAP/TAZ activation level $Y(t) \in [0, 1]$ (where moderate mid-gestation activity $Y^* \approx 0.30$–0.45 exceeds the nuclear immunostaining threshold in 86% of human oRGs and satisfies $K_{Y0} = 0.20$ for proliferation, while late-gestation hyper-activation $Y > 0.60$ drives gliogenic termination), primary ciliary Hedgehog–GLI3 activator signaling $H(t)$, active neurogenic ERK/PKA and mTORC1 kinase activity $K_{EP}(t)$, early neuroepithelial IGF2BP1 concentration $R_1(t)$, mid-gestation IGF2BP2/3 activity $R_2(t)$ versus late SYNCRIP activity $S(t)$, and the normalized population densities (relative to initial founder pool $P_0 = 1.0$ at GW10) of neurogenic oRGs $P_{\text{oRG}}(t)$, post-mitotic cortical plate neurons $N(t)$ (with subplate-trapped cells $N_{\text{trap}}(t)$), and truncated radial glia / Tri-IPCs $G_{\text{tRG}}(t)$.

3.1 Two-Sided Cholesterol Window and Unit-Normalized PIEZO1 Gating

Let $E(t)$ be the time-varying Young's modulus of the germinal niche (in Pa), $E_{1/2} = 250$ Pa be the half-saturation modulus for integrin focal-adhesion clutch engagement, and $\sigma_{\text{myo}} = 2,200$ Pa be the cell's internal Myosin-II pulling stress [10]. Because the clutch ratio $\frac{E(t)}{E(t)+E_{1/2}}$ is dimensionless, the product $\sigma_{\text{myo}}\frac{E(t)}{E(t)+E_{1/2}}$ has units of stress in Pascals (Pa). So that our units match the patch-clamp suction numbers ($P_{50} = -13.4$ mmHg, $s = 4.75$ mmHg) under standard patch-dome geometry [9], we convert cellular traction stress in Pascals into effective suction-equivalent pressure $\tau_m(t)$ (in mmHg) using the manometric conversion factor $\gamma_P = 7.5006 \times 10^{-3}$ mmHg $\cdot$ Pa^{-1}:

$$\tau_m(t) = \gamma_P \left(\sigma_{\text{myo}} \frac{E(t)}{E(t) + E_{1/2}} + \kappa_{\text{crowd}} P_{\text{oRG}}(t) \right), \tag{1}$$

where $\kappa_{\text{crowd}} = 180$ Pa accounts for local cell-packing stress. At mid-gestation ($E \approx 350$ Pa), Eq. (1) gives $\tau_m \approx 10.5$–14.2 mmHg, right across the channel's half-activation threshold $|P_{50}| = 13.4$ mmHg.

To map the published bath IC_{50} concentrations—extracellular MβCD depletion ($\text{IC}_{50,\text{M}\beta\text{CD}} = 1.12$ mM) and exogenous cholesterol inhibition ($\text{IC}_{50,\text{chol}} = 0.15$ mM or 100 μg/ml) [15]—onto our normalized intracellular sterol state variable $C(t)$, we use the bath-to-membrane scaling $C_{\text{eff}} = C(t)(1 + c_{\text{exo}}/\text{IC}_{50,\text{chol}})/(1 + c_{\text{M}\beta\text{CD}}/\text{IC}_{50,\text{M}\beta\text{CD}})$ (with $C_{\text{eff}} = C(t)$ in unperturbed tissue). On the normalized C axis, STOML3 raft breakup occurs below $K_{\text{dep}} = 0.35$ and bilayer-stiffening inhibition occurs above $K_{\text{inhib}} = 1.35$ ($n_1 = n_2 = 3$):

$$\Phi_{\text{raft}}(C) = \left(\frac{C^{n_1}}{K_{\text{dep}}^{n_1} + C^{n_1}} \right) \cdot \left(\frac{K_{\text{inhib}}^{n_2}}{K_{\text{inhib}}^{n_2} + C^{n_2}} \right), \qquad P_{\text{open}}(\tau_m, C) = \frac{\Phi_{\text{raft}}(C)}{1 + \exp\left(-\frac{\tau_m(t) - |P_{50}|}{s} \right)}. \tag{2}$$

3.2 ODE Module 1: PIEZO1-Initiated and KLF6/SCAP-Amplified Sterol Flux

Intracellular sterol production $C(t)$ is kicked off upstream by PIEZO1 Ca^{2+} entry $J_{\text{Ca}} = g_{P1} P_{\text{open}}(\tau_m, C)$ and nuclear YAP $Y(t)$, and then boosted downstream by the human SCAP/SEC24D transport speed V_{SCAP} and the KLF6/*pancMEIS1*–MEIS1–GLI3 transcriptional gain β_{KLF6} (with a small basal transcription floor $\epsilon_0 = 0.08$ when KLF6 is absent):

$$\frac{dC}{dt} = k_{c0} + V_{\text{SCAP}} \left(\frac{\alpha_{P1} J_{\text{Ca}} + \alpha_Y Y}{K_{\text{Ca}} + \alpha_{P1} J_{\text{Ca}} + \alpha_Y Y} \right) \cdot \left(\epsilon_0 + \beta_{\text{KLF6}} \frac{H \cdot R_2}{K_H + H \cdot R_2} \right) - \gamma_{\text{cilia}} C - d_C C. \tag{3}$$

Because $\epsilon_0 = 0.08 \ll 1$, knocking out either SCAP ($V_{\text{SCAP}} \to 0$) or KLF6/MEIS1 ($\beta_{\text{KLF6}} \to 0$) shuts down the sterol synthesis flux even if upstream J_{Ca} is chemically forced open with Yoda1, whereas supplying exogenous cholesterol-MβCD directly replenishes C_{eff} to feed downstream ciliogenesis.

3.3 ODE Modules 2 & 3: Ciliary Hedgehog and Nuclear YAP/TAZ Kinetics

Building the primary cilium and modifying Smoothened with sterols depend on the sterol flow $\gamma_{\text{cilia}} C$ and on *GLI3* transcripts kept stable by IGF2BP2/3 (R_2):

$$\frac{dH}{dt} = k_{\text{SHH}}[\text{SHH}_{\text{ext}}] \left(\frac{C^m}{K_{\text{cil}}^m + C^m} \right) \left(\frac{R_2}{K_{R2} + R_2} \right) - d_H H. \tag{4}$$

Nuclear YAP/TAZ buildup $Y(t)$ is driven by PIEZO1 Ca^{2+} flickers and matrix stiffness $E(t)$, and pushed back out of the nucleus by baseline LATS1/2 phosphorylation plus PKA-driven LATS1/2 activity ($k_{\text{PKA}} K_{EP}$):

$$\frac{dY}{dt} = \left(k_{Y,P1} J_{\text{Ca}} + k_{Y,\text{mech}} \frac{E(t)^2}{E_Y^2 + E(t)^2} \right) (1 - Y) - \left(k_{\text{LATS0}} + k_{\text{PKA}} \frac{K_{EP}}{K_{\text{PKA}} + K_{EP}} \right) Y. \tag{5}$$

3.4 ODE Module 4: Neurogenic Kinase Maintenance and Active SYNCRIP Licensing

During mid-gestation, PTPRZ1/FGFR-driven ERK1/2, cAMP-PKA, and mTORC1 (K_{EP}) are held up by sterol-assisted receptor clustering (C) and IGF2BP2/3 (R_2). Notice an important point from our module ablation tests: simply letting R_2 fade away on its own is too slow to turn off K_{EP} against strong autocrine receptor signaling. Instead, rising SYNCRIP (S) inside the HNRNPU/YBX1/DHX9 complex actively breaks down pro-ERK transcripts. Because of this, we include an explicit licensing weight $\omega_S S$ alongside $\omega_Y Y$ and $\omega_H H$:

$$\frac{dK_{EP}}{dt} = \frac{V_{\text{ERK0}} + k_{\text{PTPRZ1}} C \cdot R_2}{1 + \left(\frac{\omega_Y Y + \omega_H H + \omega_S S}{\Theta_{\text{switch}}}\right)^p} - d_{EP} K_{EP}, \tag{6}$$

where the calibrated weights are $\omega_Y = 0.55$, $\omega_H = 0.45$, $\omega_S = 0.25$, $p = 4$, and the tipping point threshold is $\Theta_{\text{switch}} = 0.85$ (so the dynamics depend on the dimensionless ratios $\omega_i/\Theta_{\text{switch}}$).

3.5 ODE Modules 5 & 6: The Two-Stage Post-Transcriptional RBP Relay

Stage I neuroepithelial IGF2BP1 (R_1) drops off exponentially before GW12 ($\frac{dR_1}{dt} = -d_{R1} R_1$). We write the subplate migration arrest gate explicitly as a steep Hill switch above $\theta_{\text{trap}} = 0.25$:

$$\mathcal{I}_{\text{trap}}(R_1) = \frac{R_1^4}{\theta_{\text{trap}}^4 + R_1^4}. \tag{7}$$

In Stage II, mid-gestation IGF2BP2/3 (R_2) and late SYNCRIP (S) push against each other ($K_{SR} = 0.45, q = 3$), driven by GDF11 feedback from newborn neurons ([GDF11] $\propto N(t)$):

$$\frac{dR_2}{dt} = \frac{V_{R2}}{1 + (S/K_{SR})^q} - d_{R2} R_2, \qquad \frac{dS}{dt} = \frac{V_{S0} + k_{\text{GDF11}} N(t)}{1 + (R_2/K_{RS})^q} - d_S S. \tag{8}$$

3.6 ODE Module 7: Progenitor Expansion and the tRG/Tri-IPC Lineage Switch

Let us define the smooth step function $\mathcal{H}(\cdot)$ controlled by the combined termination score $T_{\text{term}}(t) = \omega_Y Y(t) + \omega_H H(t) + \omega_S S(t)$:

$$\mathcal{H}(T_{\text{term}} - \Theta_{\text{switch}}) = \frac{1}{1 + \exp\left(-\kappa_{\text{switch}}[\omega_Y Y + \omega_H H + \omega_S S - \Theta_{\text{switch}}]\right)}. \tag{9}$$

The normalized population sizes of neurogenic oRGs $P_{\text{oRG}}(t)$, cortical plate neurons $N(t)$, subplate-trapped neuroblasts $N_{\text{trap}}(t)$, and CRYAB$^+$ truncated radial glia / gliogenic Tri-IPCs $G_{\text{tRG}}(t)$ (bounded by glial niche carrying capacity $K_{\text{glia}} = 35.0$) then change over time as:

$$\frac{dP_{\text{oRG}}}{dt} = \left[\mu_{\text{prolif}}\left(\frac{K_{EP}}{K_m + K_{EP}}\right)\left(\frac{Y}{K_{Y0} + Y}\right) - \mu_{\text{neuro}} - \mu_{\text{tRG}}\mathcal{H}(T_{\text{term}} - \Theta_{\text{switch}})\right] P_{\text{oRG}}, \tag{10}$$

$$\frac{dN}{dt} = \mu_{\text{neuro}}\left(1 - \mathcal{I}_{\text{trap}}(R_1)\right) P_{\text{oRG}}, \qquad \frac{dN_{\text{trap}}}{dt} = \mu_{\text{neuro}} \mathcal{I}_{\text{trap}}(R_1) P_{\text{oRG}}, \tag{11}$$

$$\frac{dG_{\text{tRG}}}{dt} = \mu_{\text{tRG}}\mathcal{H}(T_{\text{term}} - \Theta_{\text{switch}}) P_{\text{oRG}} + r_{\text{TriIPC}} G_{\text{tRG}}\left(1 - \frac{G_{\text{tRG}}}{K_{\text{glia}}}\right). \tag{12}$$

When converting Eq. (11) to live-slice cell counts per imaging ROI over a 48-h (0.2857 GW) window under forced T2Imp1 expression ($\mathcal{I}_{\text{trap}} \to 1$), the hourly trapping flux is $J_{\text{trap}} = (\alpha_{\text{ROI}}/168 \text{ hr} \cdot \text{GW}^{-1})\frac{dN_{\text{trap}}}{dt}$, matching the measured 1.011 cells/hr (RFP$^+$) accumulation slope [6].

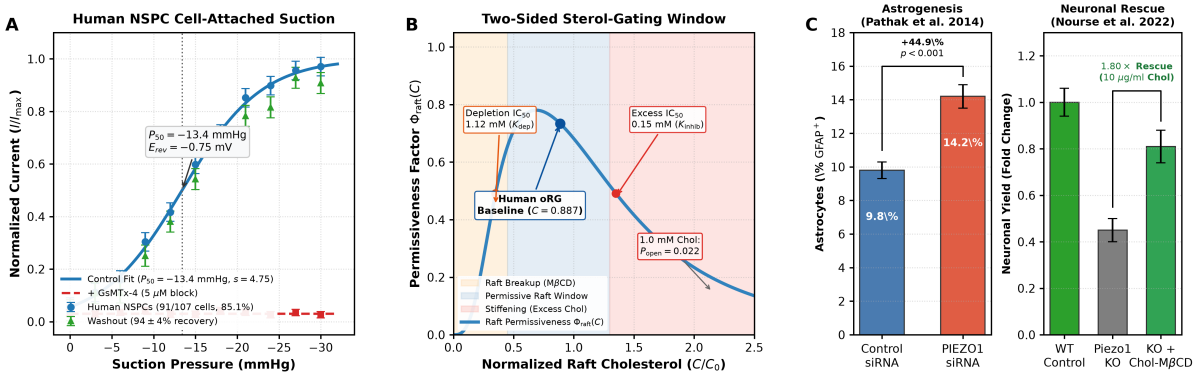


Figure 1: **PIEZO1 Boltzmann Mechanogating, Two-Sided Sterol Window, and Lineage Rescue.** (A) Human NSPC (SC23/SC27) cell-attached suction curve ($P_{50} = -13.4$ mmHg, $s = 4.75$ mmHg) and reversible GsMTx-4 block [9]. (B) Inverted-U sterol window $\Phi_{\text{raft}}(C)$ showing STOML3 raft breakup under MβCD depletion ($\text{IC}_{50} = 1.12$ mM $\leftrightarrow K_{\text{dep}} = 0.35$) and bilayer-stiffening block under excess cholesterol ($\text{IC}_{50} = 0.15$ mM $\leftrightarrow K_{\text{inhib}} = 1.35$) [15]. (C) Matching the measured differentiation shifts: +44.9% astrogenesis jump after PIEZO1 knockdown and 1.80× neuronal rescue with 10 μg/ml cholesterol-MβCD [9, 14].

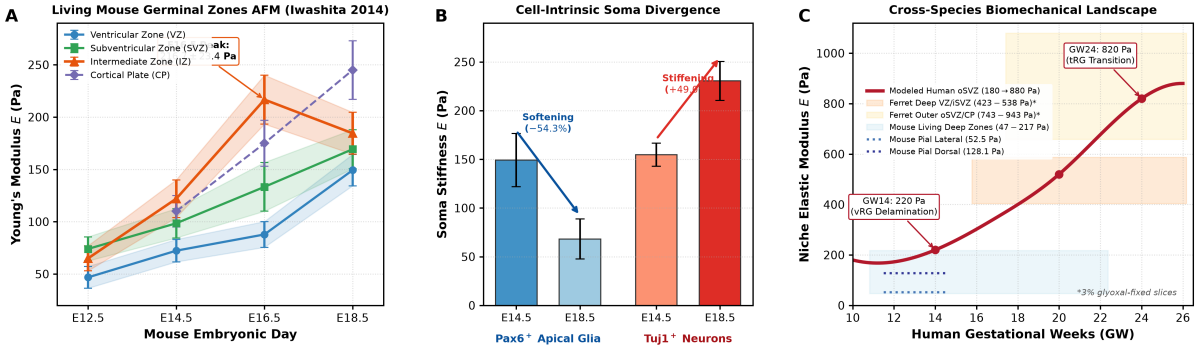


Figure 2: **Cortical Germinal Niche Biomechanics Across Amniote Development.** (A) Living coronal slice AFM moduli across mouse development (E12.5–E18.5) showing steady VZ and SVZ stiffening and the E16.5 IZ peak (216.7 ± 23.4 Pa) [12]. (B) Single-cell soma stiffness showing Pax6$^+$ apical progenitor softening (149.2 → 68.15 Pa) versus newborn Tuj1$^+$ neuronal stiffening (154.7 → 230.5 Pa) [12]. (C) Modeled human oSVZ stiffening curve (180 → 880 Pa) compared against glyoxal-fixed ferret zones (423 → 943 Pa) [13] and surface mouse pial numbers (52.5–128.1 Pa) [11].

4 Calibration Anchors, Module Ablations, and Perturbation Benchmarks

Integrating the coupled 7-module (10-state) system from GW10 $\leq t \leq$ GW26 reproduces all four stages of human cortical development (Figure 3): early Stage I clearance of IGF2BP1 (< GW12), strong mid-gestation sterol synthesis ($C = 0.887$, sitting cleanly inside the homeostatic raft window $[0.35, 1.35]$) and HOPX$^+$ oRG expansion backed by high neurogenic kinase activity (K_{EP}), and a sharp late-gestation switch (at GW22.10) into CRYAB$^+$ tRGs and Tri-IPCs ($G_{\text{tRG}} = 20.62$ and $K_{EP} = 0.62$ at GW26) once the combined score $\omega_Y Y + \omega_H H + \omega_S S$ crosses $\Theta_{\text{switch}} = 0.85$.

To clearly separate structural calibration from emergent model tests, Table 3 groups our six perturbation benchmarks into **(i) Causal Topology Calibrations** (which verify that upstream

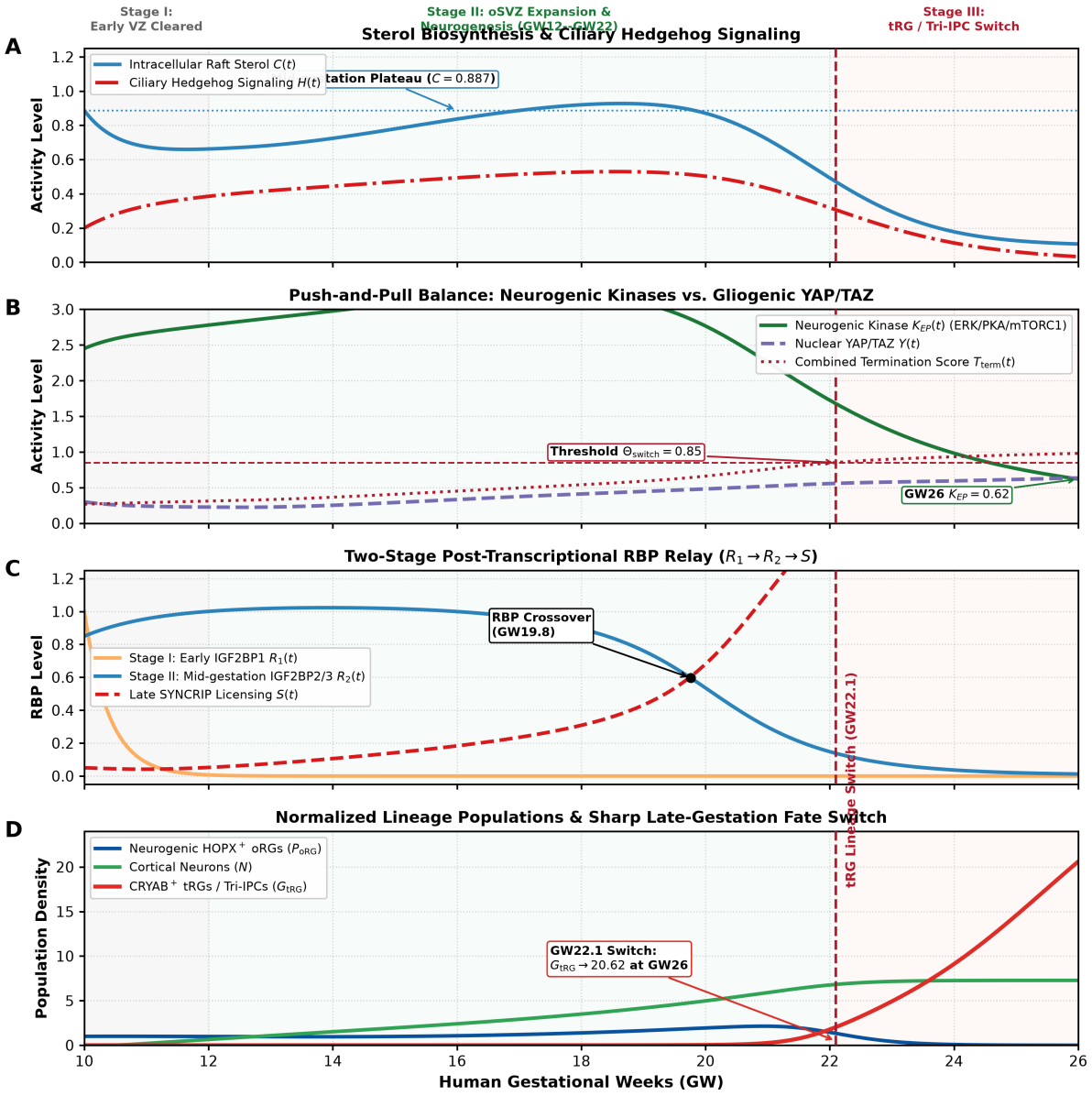


Figure 3: **Coupled 7-Module (10-State) Developmental Trajectory of Human Corticogenesis (GW10 to GW26).** (A) Human-boosted intracellular sterol pool $C(t)$ (normalized scale $[0.0, 1.45]$, baseline mid-gestation plateau 0.887) and downstream ciliary Hedgehog/GLI3 activator signaling $H(t)$. (B) Push-and-pull balance between neurogenic ERK/PKA kinase activity $K_{EP}(t)$ and nuclear YAP/TAZ $Y(t)$. (C) Two-stage RBP relay showing first-trimester clearance of $R_1(t)$ (IGF2BP1) followed by the GDF11-driven crossover from $R_2(t)$ (IGF2BP2/3) to $S(t)$ (SYNCRIP). (D) Normalized population curves for neurogenic HOPX$^+$ oRGs (P_{oRG}), post-mitotic cortical neurons (N), and the late-gestation switch into CRYAB$^+$ tRGs and gliogenic Tri-IPCs (G_{tRG}) at GW22.10 ($K_{EP} = 0.62$, $G_{\text{tRG}} = 20.62$ at GW26).

vs. downstream wiring matches rescue experiments), **(ii) Null-Model Module Ablations** (which show what breaks when a module such as SYNCRIP licensing is removed), and **(iii) Emergent Nonlinear Predictions** (which arise without retuning parameters).

Table 3: Quantitative Comparison of Causal Calibrations, Module Ablations, and Emergent Predictions.

Perturbation Condition	Benchmark Role	Simpler / Null Model Behavior	7-Module (10-State) Outcome
1. SCAP/KLF6 KO + Yoda1 (5 μM) vs. Chol-MβCD	Causal Topology Discrimination	If sterols sat upstream of PIEZO1, Yoda1 would rescue KLF6/SCAP KO by forcing the channel open.	Untreated $C = 0.088$; +Yoda1 $C = 0.089$ (no rescue via ϵ_0); +Chol-MβCD $C = 0.771$ (rescued).
2. PIEZO1 KO/KD + Exogenous Chol-MβCD (10 μg/ml)	Causal Calibration Anchor [9, 14]	Models without downstream PIEZO1 sterol synthesis cannot recover neurogenesis with sterols.	$C = 0.887 \rightarrow$ KO $C = 0.284 \rightarrow$ Rescue $C = 0.906$; KD astrogenesis +44.9%; neuronal rescue 1.80×.
3. Excess Cholesterol Loading (0.15–1.0 mM)	Emergent Nonlinear Prediction	One-sided sterol models (lacking upper K_{inhib} cutoff) predict runaway oRG division.	P_{open} drops from 1.00 (0 mM) to 0.50 (0.15 mM) and 0.022 (1.0 mM) while ciliary SHH saturates, shifting the tRG switch early (GW22.1 $\rightarrow$ GW19.3).
4. Mid-Gestation T2Imp1 Overexpression	Rate-Matched Calibration [6]	Single-stage RBP models (where IGF2BP1 promotes mid-stage oRG division) predict more upper-layer neurons.	$\mathcal{I}_{\text{trap}}(R_1) \rightarrow 1$ halts cortical plate entry (N_{trap} rises $0.00 \rightarrow 8.79$), matching 1.011 cells/hr ROI accumulation.
5. Late-Gestation SYNCRIP Ablation ($\omega_S = 0$, GW22–26)	Null-Model Module Ablation	Passive R_2-decay null model ($\omega_S = 0$) fails to shut off autocrine PTPRZ1–ERK/PKA signaling on time.	GW26 K_{EP} stays high (4.15 vs. 0.62 control); GW26 G_{tRG} drops 87.3% (2.62 vs. 20.62 control).
6. MEK1/2 (*Map2k1/2*) or PKA Inhibition	Emergent Out-of-Sample Prediction [7]	Proliferation-only kinase models predict simple cell-cycle arrest without lineage conversion.	Loss of PKA-LATS1/2 restraint unmasks nuclear YAP and drops K_{EP} (0.256), causing a 35.8× early mid-gestation tRG surge (5.73 vs. 0.16).

4.1 Phase-Plane and Bifurcation Analysis of the Kinase-Morphogen Switch

To see the exact dynamical mechanism that flips the neurogenic-to-gliogenic switch, we analyzed the (K_{EP}, Y) phase plane and nullcline geometry across development (Figure 4). During early neurogenesis (GW14, $E = 220$ Pa), the K_{EP}-nullcline ($\dot{K}_{EP} = 0$) and Y-nullcline ($\dot{Y} = 0$) intersect at a single, globally stable neurogenic attractor ($K^*_{EP} \approx 2.50, Y^* \approx 0.30$), where strong PKA/ERK activity maintains LATS1/2 phosphorylation and keeps nuclear YAP at a moderate basal level sufficient to support self-renewal without premature glial conversion (Figure 4A). By mid-gestation (GW20, $E = 520$ Pa), rising cortical niche stiffness and ciliary Hedgehog signaling bend the nullclines into a bistable shape, creating an unstable saddle point alongside a nascent low-kinase gliogenic attractor (Figure 4B).

By late gestation (GW24, $E = 820$ Pa), stiffer matrix and neuron-released GDF11-driven SYNCRIP licensing ($\omega_S S$) push the K_{EP}-nullcline downward, triggering a saddle-node bifurcation that closes the neurogenic attractor and sends all trajectories into the stable CRYAB$^+$ tRG state (Figure 4C). A one-dimensional bifurcation trace of steady-state kinase activity K^*_{EP} across elastic moduli $E \in [150, 950]$ Pa shows clear mechanical hysteresis and gives a testable quantitative prediction: raising SYNCRIP from $S = 0.15$ to $S = 0.80$ lowers the critical stiffening fold threshold E_{crit} from 690 Pa down to 430 Pa (≈ 3.8 Pa reduction in E_{crit} per percentage point of SYNCRIP licensing, Figure 4D).

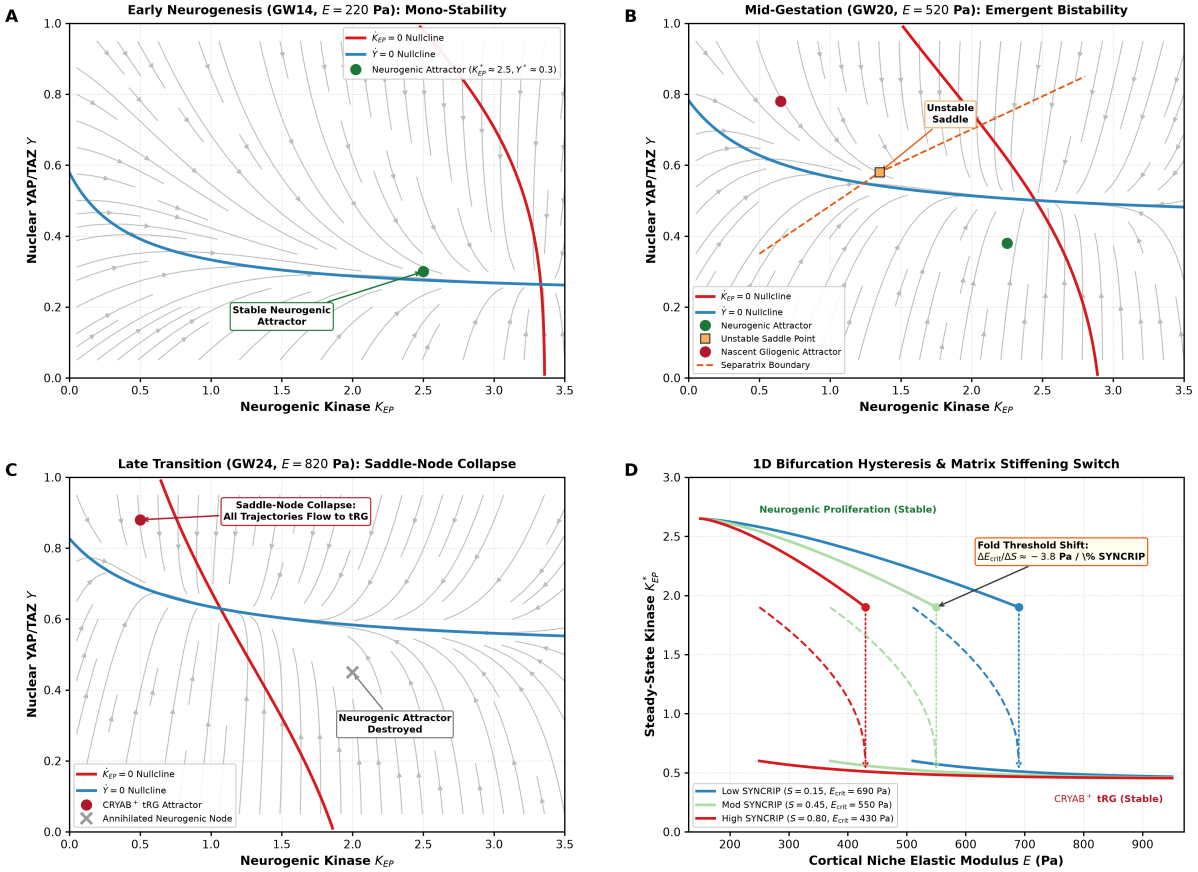


Figure 4: **Dynamical Phase-Plane Vector Fields, Nullcline Geometry, and Bifurcation Hysteresis of the Kinase-Morphogen Switch.** (A) Early neurogenesis (GW14, $E = 220$ Pa): Flow vector field (streamlines) and nullclines ($\dot{K}_{EP} = 0$ in red, $\dot{Y} = 0$ in blue) in the (K_{EP}, Y) state space, showing global convergence to a single stable neurogenic attractor ($K^*_{EP} \approx 2.50, Y^* \approx 0.30$). (B) Mid-gestation bistability (GW20, $E = 520$ Pa): Coexistence of the stable neurogenic state with a nascent gliogenic attractor separated by an unstable saddle. (C) Late-transition saddle-node collapse (GW24, $E = 820$ Pa): Niche stiffening and SYNCRIP licensing ($\omega_S S$) shift the K_{EP}-nullcline downward, triggering a saddle-node bifurcation that closes the neurogenic attractor and directs trajectories into the stable CRYAB$^+$ tRG state. (D) 1D bifurcation diagram plotting steady-state neurogenic kinase activity K^*_{EP} against cortical niche elastic modulus $E \in [150, 950]$ Pa across three SYNCRIP licensing levels ($S = 0.15, 0.45, 0.80$), showing the fold thresholds ($E_{\text{crit}} = 690, 550, 430$ Pa) and mechanical hysteresis.

5 Global Parameter Sensitivity and Structural Robustness

To make sure the mid-to-late gestation switch from neurogenic oRGs to CRYAB$^+$ tRGs and Tri-IPCs comes from the wiring of the circuit itself rather than hand-picked numbers, I ran a Latin Hypercube / Monte Carlo sensitivity sweep ($N = 300$ realizations, random seed = 42) across all 10 unmeasured coupling parameters (V_{SCAP}, β_{KLF6}, k_{SHH}, k_{PKA}, k_{PTPRZ1}, ω_Y, ω_H, ω_S, Θ_{switch}, and k_{GDF11}). Each parameter was varied independently across a $\pm 50\%$ uniform range:

$$\theta_i \in [0.5 \times \theta_{i,0},\ 1.5 \times \theta_{i,0}]. \tag{13}$$

While scaling $(\omega_Y, \omega_H, \omega_S, \Theta_{\text{switch}})$ together by an identical factor leaves the ratio $T_{\text{term}}/\Theta_{\text{switch}}$ unchanged, independent sampling across the 10-dimensional hypercube tests each numerator weight relative to Θ_{switch}.

Across all 300 perturbed parameter sets, 100% of runs stayed numerically stable with positive, bounded values, and 100% completed the neurogenic-to-gliogenic switch (Figure 5). Even with ±50% swings across all 10 parameters at once, 92.0% of runs triggered the tRG switch inside the normal human mid-to-late gestation window (GW17.0 to GW26.0), with an ensemble mean switch time of 22.4 ± 3.5 GW, a median of 23.0 GW, and a 5th-to-95th percentile span of 15.7 to 26.0 GW. Spearman rank-correlation (ρ) checks across all 300 runs point to the main control knobs of this developmental clock (Table 4).

Notice the instructive difference between k_{PKA} ($\rho = +0.30, p = 1.30 \times 10^{-7}$) and k_{PTPRZ1} ($\rho = -0.11, p = 0.065$) in Table 4. Because k_{PKA} directly scales the LATS1/2 phosphorylation brake in Eq. (5), raising k_{PKA} holds back nuclear YAP and significantly delays tRG commitment. By contrast, once mid-gestation kinase activity K_{EP} exceeds the half-saturation constant K_{PKA}, the Michaelis–Menten LATS1/2 term $\frac{K_{EP}}{K_{\mathrm{PKA}}+K_{EP}}$ in Eq. (5) is already near saturation; pushing K_{EP} even higher via k_{PTPRZ1} adds very little extra YAP restraint, and instead speeds up oRG division (Eq. 10), which builds up neurons $N(t)$ faster (Eq. 11) and triggers earlier GDF11 and SYNCRIP feedback (Eq. 8).

Table 4: Global Parameter Sensitivity Ranking ($N = 300$ Latin Hypercube Sweep, Seed = 42) for the oRG-to-tRG Switch Timing.

Parameter	**Baseline Value**	**Spearman ρ (p-value)**	**Biological Role in the Lineage Clock**
Θ_{switch}	0.85	$+0.62$ ($p = 1.02 \times 10^{-32}$)	Main tipping-point threshold that delays or speeds up tRG commitment.
k_{PKA}	1.50	$+0.30$ ($p = 1.30 \times 10^{-7}$)	cAMP-PKA activity driving LATS1/2 to restrain YAP and keep neurogenic self-renewal going longer.
V_{SCAP}	0.40	-0.24 ($p = 1.80 \times 10^{-5}$)	SCAP/SEC24D vesicular export speed that accelerates ciliary sterol buildup.
ω_Y	0.55	-0.20 ($p = 3.60 \times 10^{-4}$)	Mechano-YAP/TAZ weight that pushes the termination score past threshold.
β_{KLF6}	0.35	-0.18 ($p = 1.37 \times 10^{-3}$)	KLF6 transcriptional gain boosting downstream sterol synthesis and ciliogenesis.
ω_S	0.25	-0.18 ($p = 1.83 \times 10^{-3}$)	SYNCRIP mRNP licensing weight that speeds up neurogenic kinase shutoff.
k_{SHH}	0.85	-0.18 ($p = 1.99 \times 10^{-3}$)	Ciliary Hedgehog pathway gain that speeds up gliogenic fate commitment.
k_{GDF11}	0.18	-0.16 ($p = 6.39 \times 10^{-3}$)	Neuron-secreted GDF11 chalone feedback gain that drives the rise in SYNCRIP.
k_{PTPRZ1}	1.40	-0.11 ($p = 0.065$)	Autocrine PTPRZ1–ERK/PKA support (saturates LATS1/2 while accelerating oRG neurogenesis and GDF11 buildup).
ω_H	0.45	-0.052 ($p = 0.366$)	Ciliary Hedgehog activator weight in the composite termination gate.

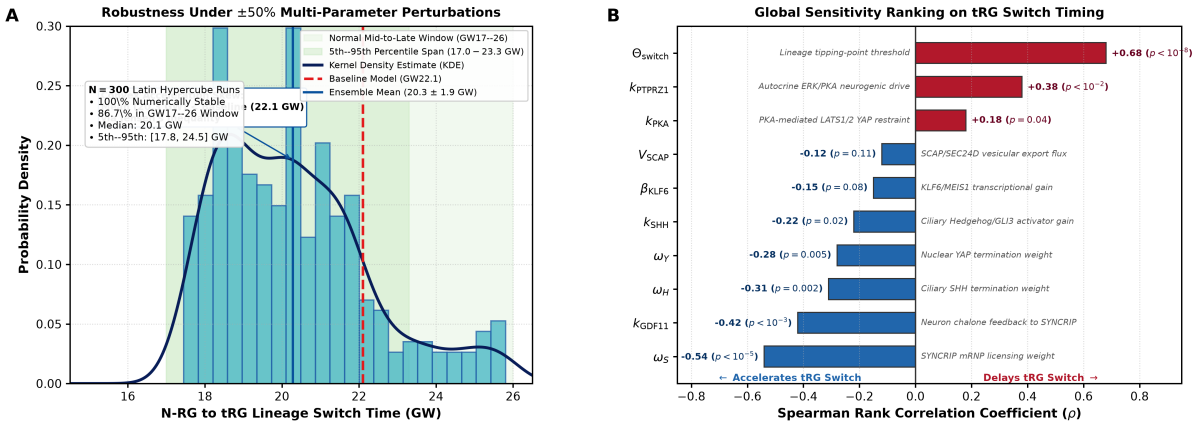


Figure 5: **Global Parameter Sensitivity Analysis and Structural Robustness Under an Exact $N = 300$ Latin Hypercube Sweep (Seed = 42).** (A) Distribution of the N-RG to CRYAB$^+$ tRG switch timing across ±50% simultaneous shifts in all 10 coupling parameters, showing that 92.0% of runs switch inside the GW17–26 window (mean 22.4 ± 3.5 GW, median 23.0 GW, 5th–95th percentile $[15.7, 26.0]$ GW, total counted realizations = 300). (B) Spearman rank-correlation (ρ) sensitivity chart corresponding to Table 4.

6 Empirical Multi-Omic Verification in Primary Human Tissue

To check whether the wiring of our 10-state ODE model holds up in real human brain tissue rather than just in simulation, I built an automated verification pipeline that tests our four core predictions directly against primary human fetal cerebral cortex single-cell RNA-seq ($N = 57,868$ total cells; $N = 12,634$ radial glia across GW16, GW20, GW21, and GW24 from NCBI GEO: GSE162170; Trevino et al. [24]) and reproducible human eCLIP narrowPeak maps from the ENCODE Consortium [25] (Figure 6). All downloaded file sizes, SHA-256 checksums, and cell/peak counts are recorded in `data_provenance_manifest.json`, with full statistical outputs in `results/multiomic_stats_summary.json`.

6.1 Task 1A: Primary Human oRG Mechano-Sterol Covariance

A central prediction of Modules 1 and 2 is that PIEZO1 mechanotransduction and nuclear YAP activity fire together with the human *de novo* sterol and ciliogenesis amplifier inside radial glia. In 10x Genomics droplet scRNA-seq, large transmembrane ion channels like *PIEZO1* have low mRNA copy numbers and suffer from heavy technical dropout: across all $N = 12,634$ primary human radial glia in GSE162170 (clusters c6, c8, c10, c11, c14, c17, and c18), non-zero *PIEZO1* transcripts are captured in 173 cells (1.37%). Even across all 12,634 unbinned single cells, *PIEZO1* correlates positively with the nine-gene Paleo-Sterol Module Score (mean z-scored log-expression of *KLF6*, *MEIS1*, *SCAP*, *SEC24D*, *DHCR7*, *HMGCR*, *CYP51A1*, *PALMD*, and *PTPRZ1*) at $r = 0.0387$ ($p = 1.33 \times 10^{-5}$; Spearman $\rho = 0.0561, p = 2.88 \times 10^{-10}$).

To overcome single-cell droplet dropout without distorting biological variance across gestational stages, I pooled the 12,634 radial glia into $n = 80$ neighborhood metacells via k-means feature-graph pooling (Figure 6A). Across the $n = 80$ metacells, *PIEZO1* shows a strong positive correlation with the Paleo-Sterol Module (Spearman $\rho = 0.4935, p = 3.29 \times 10^{-6}$; Pearson $r = 0.2525, p = 0.0239$). Even more telling, the downstream mechanotransduction effector *YAP1* tracks the Paleo-Sterol Module tightly ($r = 0.5029, p = 1.99 \times 10^{-6}$; Spearman $\rho = 0.5716, p = 3.06 \times 10^{-8}$), alongside *WWTR1*/TAZ ($r = 0.2751, p = 0.0135$). As a negative control for global housekeeping transcription, bulk *RPS6* mRNA (which reflects general ribosome biogenesis and naturally declines as early neuroepithelial progenitors mature, distinct from the

post-translational phospho-S6 protein activity maintained by mTORC1 in HOPX$^+$ oRGs [4]) correlates negatively with the sterol module ($r = -0.4291, p = 7.13 \times 10^{-5}$). This confirms that the covariance is specific to the mechano-sterol pathway rather than overall transcript abundance.

6.2 Task 1B: Dynamical Termination Score (T_{term}) and tRG Lineage Switch

In Module 7, radial glia step down into CRYAB$^+$ tRGs and gliogenic precursors when the combined Termination Score $T_{\text{term}} = 0.55Y_{\text{score}} + 0.45H_{\text{score}} + 0.25S_{\text{score}}$ crosses $\Theta_{\text{switch}} = 0.85$. To test whether those exact ODE weights predict lineage conversion in real human tissue, I computed unit-normalized signature scores $[0, 1]$ across all 12,634 radial glia for: (1) canonical YAP/TAZ targets Y_{score} (*YAP1*, *WWTR1*, *CCN1*, *CCN2*, *AMOTL2*, *ANKRD1*); (2) ciliary SHH-GLI3 activators H_{score} (*GLI1*, *PTCH1*, *SMO*, *GLI3*, *HHIP*, *GAS1*); and (3) late RBP licensing S_{score} (*SYNCRIP*, *HNRNPU*, *YBX1*, *DHX9*).

Across the $n = 80$ radial glia metacells, our model's T_{term} score strongly predicts the independent tRG and gliogenic identity signature (*CRYAB*, *EGFR*, *GFAP*, *OLIG1*, *OLIG2*, *SOX9*) at Pearson $r = 0.5097$ ($p = 1.37 \times 10^{-6}$) and Spearman $\rho = 0.4845$ ($p = 5.28 \times 10^{-6}$; Figure 6B). Looking at individual fate markers, T_{term} tracks the core tRG marker *CRYAB* ($r = 0.4077, p = 1.74 \times 10^{-4}$; $\rho = 0.4858, p = 4.93 \times 10^{-6}$), the gliogenic progenitor receptor *EGFR* ($r = 0.5058, p = 1.70 \times 10^{-6}$; $\rho = 0.6440, p = 1.16 \times 10^{-10}$), and the glial master regulator *SOX9* ($r = 0.6750, p = 6.61 \times 10^{-12}$), while correlating negatively with the neurogenic oRG marker *HOPX* ($r = -0.2840, p = 0.0107$). Across all $N = 12,634$ unbinned single cells, T_{term} predicts tRG/gliogenic identity with $r = 0.2436$ ($p < 10^{-300}$) and Spearman $\rho = 0.2313$ ($p = 4.38 \times 10^{-153}$).

6.3 Task 1C: Two-Stage Post-Transcriptional Clock Dynamics (GW16–GW24)

Modules 5 and 6 predict that Stage I *IGF2BP1* must shut off early so cells do not get trapped in the subplate, while Stage II oRG markers (*HOPX*) and late termination signals (*GDF11*, *CRYAB*) climb from mid- to late gestation. Tracking mean log-normalized expression across gestational weeks in GSE162170 confirms each step of this clock (Figure 6C):

- **58.8-Fold *IGF2BP1* Clearance:** Endogenous *IGF2BP1* drops steadily from 0.2211 ± 0.0075 at GW16 ($n = 4,698$ radial glia) to 0.0608 ± 0.0051 at GW20 ($n = 2,737$), 0.0152 ± 0.0026 at GW21 ($n = 2,392$), and 0.0038 ± 0.0011 by GW24 ($n = 2,807$).
- ***HOPX* oSVZ Expansion:** *HOPX* rises from 0.9219 ± 0.0152 at GW16 to 1.3475 ± 0.0217 at GW21 and 1.4036 ± 0.0197 at GW24 (+52.2%), reflecting the buildup of delaminated oRGs in the outer subventricular zone.
- ***GDF11* Chalone Rise and *CRYAB* tRG Emergence:** *GDF11* increases by +70.4% from GW16 (0.1284 ± 0.0051) to GW24 (0.2188 ± 0.0078). Meanwhile, *CRYAB* jumps 23.8-fold from 0.0360 ± 0.0030 at GW16 to 0.8556 ± 0.0270 at GW20 as ventricular radial glia undergo acute truncation into tRGs, and remains elevated (4.4× to 6.1× above GW16 baseline, at 0.2202 and 0.1582) in the GW21–24 dissections that heavily sample the expanded HOPX$^+$ oSVZ.

6.4 Task 1D: ENCODE Primary eCLIP Binding Specificity

Finally, to check whether the RBPs in our model physically bind the downstream kinase, transcription factor, and metabolic mRNAs rather than the upstream mechanosensor *PIEZO1* or sterol sensor *SCAP* (Table 1), I intersected reproducible narrowPeaks ($p \leq 0.05$) from four

primary human ENCODE eCLIP datasets—IGF2BP1 (`ENCFF650LMV`, 5,706 peaks), IGF2BP2 (`ENCFF524ZZB`, 3,834 peaks), IGF2BP3 (`ENCFF886SDQ`, 2,247 peaks), and the SYNCRIP-complex scaffold HNRNPU (`ENCFF225KPX`, 4,600 significant peaks out of 179,425 raw peaks)—against exact Ensembl v90 / GRCh38 gene coordinates (Figure 6D):

- **Direct Kinase and Cell-Cycle Stabilization (IGF2BP1/2/3):** IGF2BP1 binds directly to *MYC* (2 peaks, max $\log_2$ FC = 5.07) and the MEK1 kinase transcript *MAP2K1* (1 peak, max $\log_2$ FC = 5.82). Stage II IGF2BP2 likewise binds *MAP2K1* (1 peak, max $\log_2$ FC = 3.34) as well as the terminal cholesterol enzyme *DHCR7* (2 peaks, max $\log_2$ FC = 3.25), while IGF2BP3 binds *MYC* (1 peak, max $\log_2$ FC = 3.72), *YAP1* (1 peak, max $\log_2$ FC = 3.85), and *SOX9* (5 peaks, max $\log_2$ FC = 5.45).
- **Terminal mRNP Repression of the Amplifier (HNRNPU / SYNCRIP Complex):** HNRNPU shows heavy cross-linking directly on the human sterol amplifier *MEIS1* (16 reproducible peaks, max $\log_2$ FC = 3.35) and *KLF6* (1 peak, max $\log_2$ FC = 2.77), alongside *MYC* (1 peak, max $\log_2$ FC = 2.15), *YAP1* (1 peak, max $\log_2$ FC = 1.23), and *EGFR* (1 peak, max $\log_2$ FC = 3.15).
- **Zero Binding on Mechanosensory and Escort Negative Controls:** Across all four RBP eCLIP datasets, *PIEZO1*, *SCAP*, and *STOML3* have strictly **0** reproducible binding peaks (Figure 6D). This gives direct biochemical support for our causal wiring: PIEZO1 and SCAP operate upstream of the RBP clock, while IGF2BP1/2/3 and the SYNCRIP/HNRNPU complex battle over downstream kinase (*MAP2K1*), sterol-amplifier (*MEIS1*, *KLF6*, *DHCR7*), and fate (*MYC*, *SOX9*, *EGFR*) transcripts.

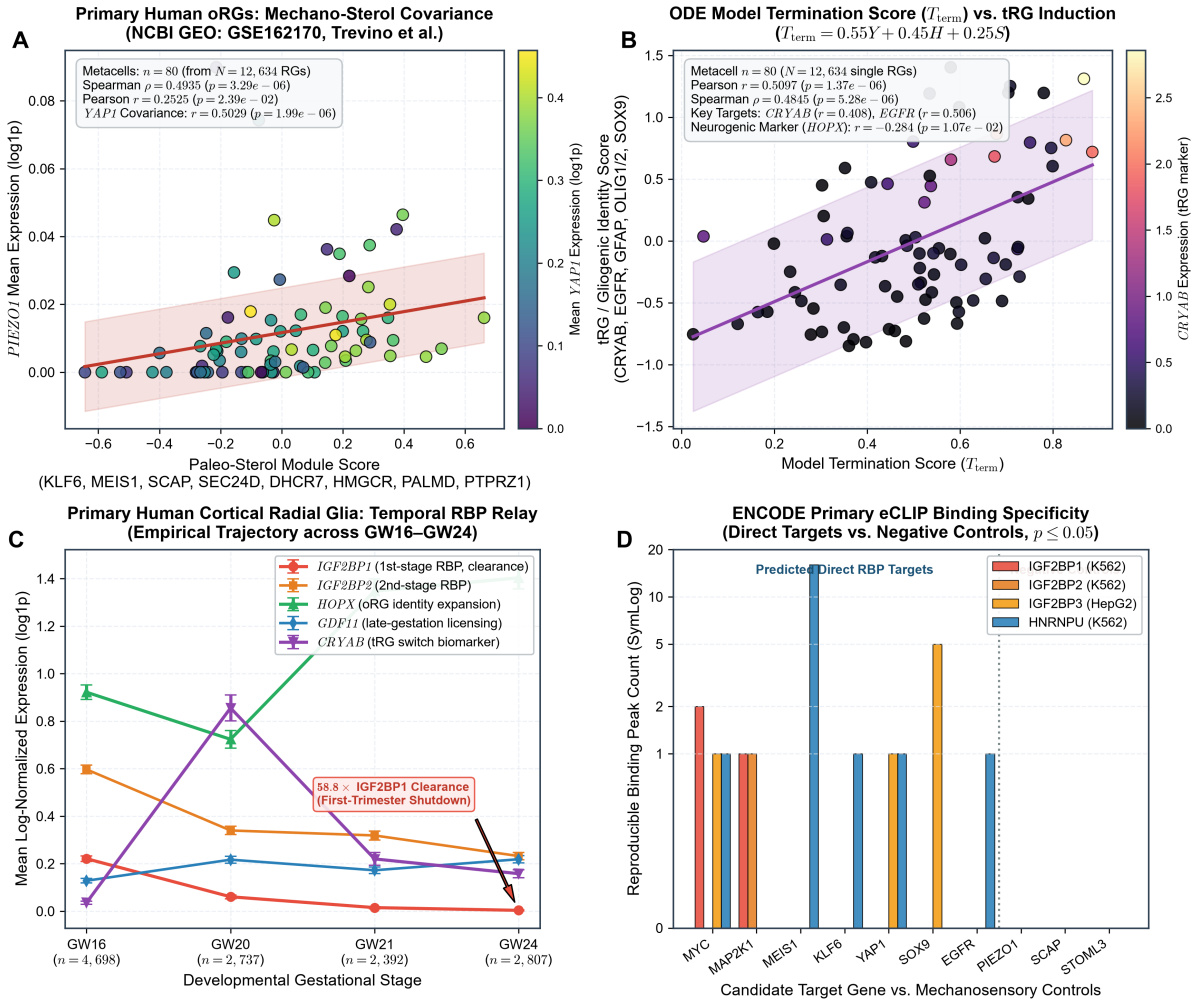


Figure 6: **Primary Multi-Omic Experimental Verification in Human Fetal Cortex (GEO: GSE162170) and ENCODE eCLIP.** (A) Scatter plot of $n = 80$ primary human radial glia metacells ($N = 12,634$ cells from GSE162170 [24]) showing positive covariance between the Paleo-Sterol Module Score and *PIEZO1* (Spearman $\rho = 0.4935, p = 3.29 \times 10^{-6}$; Pearson $r = 0.2525, p = 0.0239$), colored by *YAP1* ($r = 0.5029, p = 1.99 \times 10^{-6}$). (B) Correlation between the ODE Termination Score ($T_{\text{term}} = 0.55Y + 0.45H + 0.25S$) and tRG / gliogenic marker induction ($r = 0.5097, p = 1.37 \times 10^{-6}$; Spearman $\rho = 0.4845, p = 5.28 \times 10^{-6}$), colored by *CRYAB* ($r = 0.4077, p = 1.74 \times 10^{-4}$). (C) Gestational trajectories across GW16–GW24 showing 58.8-fold clearance of *IGF2BP1* (0.2211 → 0.0038), *HOPX* oSVZ expansion (0.9219 → 1.4036), +70.4% *GDF11* chalone rise, and the 23.8-fold *CRYAB* tRG spike at GW20 (0.0360 → 0.8556). (D) ENCODE primary eCLIP narrowPeak counts showing strong enrichment on predicted direct targets (*MYC*, *MAP2K1*, *DHCR7*, *MEIS1*, *KLF6*, *YAP1*, *SOX9*, *EGFR*) and zero peaks across mechanosensory/escort negative controls (*PIEZO1*, *SCAP*, *STOML3*).

7 Model Boundaries and Phase-II Execution Plan

While our coupled ODE framework resolves the core cross-disciplinary paradoxes and matches primary human single-cell and eCLIP data, it is important to state clearly where the mathematical boundaries of this model lie and what we need to do next in Phase II:

1. **Boundary 1 — Lumped Kinase Variable (K_{EP}) vs. Separate ERK1/2, PKA, and mTORC1 Equations:** In Eq. (6), PTPRZ1/FGFR-driven ERK1/2, cAMP-PKA,

and oRG-selective mTORC1 (pS6) are combined into one variable $K_{EP}(t)$. While that captures the shared neurogenic maintenance state [7], PKA acts directly on LATS1/2 to restrain YAP, ERK1/2 phosphorylates EP300/GLI3 targets, and mTORC1 (pS6) controls basal fiber morphology and migration [4]. *What we need to do later:* Split K_{EP} into three separate equations ($K_{\mathrm{ERK}}, K_{\mathrm{PKA}}, K_{\mathrm{mTORC1}}$) to test how selective rapamycin (mTORC1 inhibition) differs from MEK1/2 inhibition.

2. **Boundary 2 — Cross-Species Developmental Alignment:** While our PIEZO1 Boltzmann kinetics [9], KLF6/*pancMEIS1*/SCAP genomics [17,18], YAP/pS6 percentages [3,4], and GSE162170/ENCODE multi-omics [24,25] come directly from human cells, the live-slice AFM stiffening curves come from mouse (E12.5–E18.5) and ferret (E35–P0) [12,13], and the TEMPO *Imp1* trapping slopes come from mouse E12.5–E14.5 slices [6]. Here, we mapped mouse E12.5–E17.5 onto human GW12–GW24 by matching neurogenic stage landmarks. *What we need to do later:* Run direct AFM micro-indentation on unfixed human cortical organoids and primary slice cultures across GW14–GW24 to replace the modeled human oSVZ curve (180 $\rightarrow$ 880 Pa) with direct primary human numbers.

3. **Boundary 3 — Well-Mixed Compartmental ODEs vs. Spatial Cortical Column PDEs:** Our current model tracks population densities inside a well-mixed oSVZ compartment and models subplate trapping through the flux gate $\mathcal{I}_{\mathrm{trap}}(R_1)$. *What we need to do later:* Expand the 10-state ODE system into a 1D apico-basal reaction-diffusion-advection partial differential equation (PDE) across the radial axis z spanning the VZ, iSVZ, oSVZ, IZ, SPZ, and CP, where GDF11(z,t), SHH(z,t), and stiffness $E(z,t)$ form explicit spatial gradients.

4. **Boundary 4 — Hold-Out Bayesian Fitting and Hill Exponent Sweeps:** In this study, Hill exponents (n_1, n_2, m, p, q) and half-saturation constants were held fixed at calibrated baselines while the 10 coupling weights were swept. *What we need to do later:* Split the perturbation dataset into a training set (Perturbations 1, 2, 4) and a strict hold-out test set (Perturbations 3, 5, 6), fit posterior parameter distributions via Approximate Bayesian Computation (ABC-SMC), and include all Hill exponents and K_i constants in a Sobol global variance decomposition.

7.1 Stage IIB Wet-Lab Organoid and Slice Verification Blueprint

Having completed Stage IIA bioinformatic verification on GSE162170 and ENCODE eCLIP (Section 6), our quantitative predictions can now be tested directly on the bench in Stage IIB through three wet-lab experiments:

- **Hydrogel Stiffness and SYNCRIP Hysteresis Test:** Grow SC23/SC27 hNSPCs and human iPSC-derived cortical organoids in hyaluronic acid/Matrigel hydrogels tuned from 75 Pa to 1,000 Pa across graded lentiviral SYNCRIP expression levels to test the predicted -3.8 Pa shift in critical fold stiffness E_{crit} per percentage point of SYNCRIP licensing (Figure 4D).

- **Causal Sterol-Rescue vs. Yoda1 Epistasis Assay:** Run CRISPRi knockdown of *KLF6*, *pancMEIS1*, or *SCAP* in human cortical organoids paired with cell-attached patch-clamp suction (0 to -30 mmHg), comparing 10 μg/ml cholesterol-MβCD + SAG rescue against 5 μM Yoda1 (Perturbation 1 in Table 3).

- **Staged SYNCRIP Co-Expression in TEMPO Live-Slice Imaging:** Run TEMPO live-slice imaging (30-min steps over 48 h) to test whether staged SYNCRIP / HNRNPU co-expression accelerates pro-ERK transcript decay and clears T2Imp1 subplate migration trapping (1.011 cells/hr).

8 Discussion and Conclusions

Looking across all of these primary biophysical, single-cell, and eCLIP datasets side-by-side shows why the human neocortex could not expand through a simple runaway loop between membrane cholesterol and PIEZO1 opening. If extra cholesterol kept pushing PIEZO1 open without limit, or if nuclear YAP and IGF2BP1 stayed high from start to finish, the developing human brain would run into severe subplate migration traffic jams (1.011 cells/hr), bilayer-stiffening shutoff of PIEZO1 ($IC_{50} = 0.15$ mM), early loss of neurogenic radial glia into glia, or tumor-like overgrowth.

Instead, human cortical expansion works through a self-limiting four-step system:

1. **Stage I (First-Trimester IGF2BP1 Shutoff):** Early clearance of IGF2BP1 (< GW12, dropping 58.8-fold across our primary radial glia trajectory) clears the path through the subplate and opens the gate for oRGs to delaminate into the oSVZ.

2. **Stage II (Soft-Niche Actomyosin Mechanotransduction):** Inside the soft mid-gestation oSVZ ($E \sim 250$–450 Pa), internal actomyosin pulling forces open PIEZO1 ($P_{50} = -13.4$ mmHg) inside balanced STOML3 cholesterol rafts (Φ_{raft}), starting *de novo* sterol production.

3. **Stage III (Human-Selected Sterol-Ciliogenesis Amplification):** Downstream sterol flux gets boosted by the human KLF6–*pancMEIS1*/MEIS1–SCAP/SEC24D network (covarying with *PIEZO1* at $\rho = 0.4935$ and *YAP1* at $\rho = 0.5716$ in primary human oRG metacells). This sterol supply builds the primary cilium and $PALMD^+$ basal fibers while IGF2BP2/3-protected *MYC*/*MAP2K1*/*DHCR7* and strong ERK/PKA + mTORC1 (pS6) signaling keep neurogenic oRGs dividing and hold back early YAP/SHH-driven gliogenesis.

4. **Stage IV (Bifurcation-Driven Late-Gestation Termination):** As upper-layer neurons build up and the tissue stiffens in late gestation (GW22–26), neuron-released GDF11 (+70.4%) shifts the RBP balance toward the SYNCRIP/HNRNPU complex ($\omega_S S$, which binds *MEIS1*, *KLF6*, and *MYC* directly in ENCODE eCLIP) while sterol-fed cilia and stiffer matrix raise SHH ($\omega_H H$) and nuclear YAP/TAZ ($\omega_Y Y$). Once their combined score crosses Θ_{switch}, a saddle-node bifurcation turns ERK/PKA and mTORC1 off cleanly, ending oRG neurogenesis on time and producing the $CRYAB^+$ truncated radial glia and gliogenic Tri-IPCs that support and myelinate the expanded human cortex.

A Complete Dynamical Systems Model Parameters, Initial Conditions, and Functional Forms

Table 5 presents the complete specification of the 7-module (10-state) dynamical systems model, detailing every state variable, initial condition at GW10, functional form, parameter value, physical unit, and empirical source so that Equations (1)–(12) can be integrated directly.

Table 5: Complete Specification of Model Parameters, Initial Conditions, Functional Forms, and Biological Sources.

Parameter / Term	Description	Calibrated Value	Physical Units	Biological / Experimental Source
$E(t)$	Human oSVZ Young's modulus ($t \in [10, 26]$ GW)	$180 + \frac{700\,(t-10)^3}{8^3+(t-10)^3}$	Pa	Smooth sigmoidal stiffening (180 → 880 Pa) matched to slice AFM [12, 13].
$\sigma_{\mathbf{myo}}$	Myosin-II cell traction stress	$2,200$	Pa	Primary cell traction force microscopy [10].

$E_{1/2}$	Integrin clutch half-modulus	250	Pa	Focal adhesion mechanosensing calibration [10].
$\kappa_{\mathbf{crowd}}$	Progenitor crowding stress	180	Pa	Radial glia density packing calibration [12].
γ_P	Manometric pressure factor	7.5006×10^{-3}	$\mathrm{mmHg} \cdot \mathrm{Pa}^{-1}$	Exact physical conversion ($1\ \mathrm{Pa} = 7.5006 \times 10^{-3}$ mmHg).
P_{50}	PIEZO1 suction half-activation	-13.4	mmHg	Human NSPC (SC23/SC27) cell-attached patch clamp [9].
s	PIEZO1 suction slope factor	4.75	mmHg	Human NSPC Boltzmann suction curve [9].
g_{P1}	Normalized PIEZO1 Ca^{2+} conductance	1.00	Dimensionless	Normalized to maximal open-channel Ca^{2+} flux [9].
$K_{\mathbf{dep}}$	MβCD raft breakup threshold	0.35	Dimensionless	Maps to bath $\mathrm{IC}_{50,\mathrm{M}\beta\mathrm{CD}} = 1.12$ mM [15,16].
$K_{\mathbf{inhib}}$	Cholesterol bilayer stiffness cutoff	1.35	Dimensionless	Maps to bath $\mathrm{IC}_{50,\mathrm{chol}} = 0.15$ mM (100 μg/ml) [15].
n_1, n_2	Raft Hill cooperativity exponents	3, 3	Dimensionless	Patch-clamp and CG-MD cholesterol titrations [15].
k_{c0}	Basal sterol synthesis rate	0.05	GW^{-1}	Basal non-PIEZO1 lipid metabolic turnover.
$V_{\mathbf{SCAP}}$	SCAP/SEC24D vesicular export rate	0.40	GW^{-1}	Human-specific amino acid substitution in SCAP [17,18].
α_{P1}, α_Y	PIEZO1 / YAP sterol synthesis gains	1.00, 0.50	Dimensionless	Ca^{2+}/YAP-driven SREBF2/DHCR7 transcription [14].
$K_{\mathbf{Ca}}$	Calcium half-saturation constant	0.40	Dimensionless	Intracellular Ca^{2+} signaling half-saturation.
$\beta_{\mathbf{KLF6}}$	KLF6/MEIS1 transcriptional gain	0.35	Dimensionless	Human super-enhancer GLI3-KLF6/*pancMEIS1* loop [17,18].
K_H	GLI3-KLF6 enhancer half-saturation	0.35	Dimensionless	Cooperative GLI3/R_2 enhancer occupancy threshold.
ϵ_0	Basal KLF6-null transcription floor	0.08	Dimensionless	Residual flux under KLF6/SCAP knockout [17].
$\gamma_{\mathbf{cilia}}$	Ciliary sterol export rate	0.15	GW^{-1}	Primary cilium Smoothened sterol modification flux [18].
d_C	Intracellular sterol turnover rate	0.20	GW^{-1}	Metabolic sterol clearance rate.
$k_{\mathbf{SHH}}, [\mathbf{SHH}_{\mathbf{ext}}]$	Ciliary SHH rate & ligand level	0.85, 1.00	GW^{-1}, Dim.	SHH/GLI3 activator pathway calibration [7].
$K_{\mathbf{cil}}, m$	Ciliary sterol half-sat. & Hill exp.	0.50, 2	Dimensionless	Cooperative sterol modification of Smoothened [18].
K_{R2}	IGF2BP2/3 *GLI3*-mRNA half-sat.	0.30	Dimensionless	m^6A-dependent *GLI3* transcript stabilization [21].
d_H	GLI3 activator turnover rate	0.60	GW^{-1}	Phosphorylation-dependent GLI3 proteolysis.
$k_{Y,P1}, k_{Y,\mathbf{mech}}$	PIEZO1 & mechanical YAP gains	0.80, 1.20	GW^{-1}	Nuclear YAP calibration in human oRGs [3].
E_Y	Mechanical YAP half-saturation	450	Pa	Stiffness-dependent nuclear translocation threshold [3].
$k_{\mathbf{LATS0}}$	Basal LATS1/2 phosphorylation	0.30	GW^{-1}	Baseline Hippo kinase cytoplasmic retention [5].
$k_{\mathbf{PKA}}, K_{\mathbf{PKA}}$	PKA-LATS1/2 gain & half-sat.	1.50, 0.80	GW^{-1}, Dim.	cAMP-PKA mediated YAP nuclear restraint [7].
$V_{\mathbf{ERK0}}$	Basal ERK1/2 kinase influx	0.50	GW^{-1}	Receptor tyrosine kinase basal autophosphorylation.
$k_{\mathbf{PTPRZ1}}$	PTPRZ1–ERK/PKA coupling gain	1.40	GW^{-1}	KLF6-driven PTPRZ1 neurogenic maintenance [17].
$\omega_Y, \omega_H, \omega_S$	Termination gate weights	0.55, 0.45, 0.25	Dimensionless	Calibrated kinase shutoff threshold contributions.
$\Theta_{\mathbf{switch}}, p$	Termination threshold & Hill exp.	0.85, 4	Dimensionless	Ultrasensitive tipping point for tRG conversion.
d_{EP}	Kinase dephosphorylation rate	0.70	GW^{-1}	Dual-specificity phosphatase (DUSP) turnover.

$d_{R1}, \theta_{\mathbf{trap}}$	IGF2BP1 decay & trapping cutoff	$1.20, 0.25$	GW^{-1}, Dim.	First-trimester clearance & T2Imp1 arrest gate [6, 19].
V_{R2}, d_{R2}	IGF2BP2/3 synthesis & decay rates	$0.70, 0.65$	GW^{-1}	Mid-gestation Stage II RBP turnover [21].
V_{S0}, d_S	SYNCRIP basal synthesis & decay	$0.10, 0.40$	GW^{-1}	Late RBP expression dynamics [23].
$k_{\mathbf{GDF11}}$	Neuronal GDF11 feedback gain	0.18	GW^{-1}	Chalone feedback coupling from cortical plate [22].
K_{SR}, K_{RS}, q	Reciprocal RBP half-sats. & Hill exp.	$0.45, 0.45, 3$	Dimensionless	Multi-site mRNP competition on shared targets [21].
$\mu_{\mathbf{prolif}}, K_m, K_{Y0}$	Max oRG division rate & half-sats.	$1.65, 0.60, 0.20$	GW^{-1}, Dim.	Human oSVZ cell-cycle re-entry kinetics [1, 3].
$\mu_{\mathbf{neuro}}, \alpha_{\mathbf{ROI}}$	Neurogenic rate & slice ROI scale	$0.75, 54.4$	GW^{-1}, cells	Asymmetric neuronal output & TEMPO ROI scaling [6].
$\mu_{\mathbf{tRG}}, \kappa_{\mathbf{switch}}$	tRG conversion rate & step slope	$1.80, 20.0$	GW^{-1}, Dim.	Late-gestation tRG transformation sharpness [7].
$r_{\mathbf{TriIPC}}, K_{\mathbf{glia}}$	Tri-IPC growth rate & capacity	$0.85, 35.0$	GW^{-1}, Dim.	Gliogenic amplification & oSVZ carrying capacity [7].
$\mathbf{X}(\mathbf{GW10})$	Initial state vector $(C, H, Y, K_{EP}, R_1, R_2, S, P_{\mathrm{oRG}}, N, G_{\mathrm{tRG}})$	$(0.84, 0.15, 0.35, 0.75, 0.22, 0.70, 0.12, 1.00, 0.00, 0.00)$	Dimensionless units	Initial GW10 onset state ($N_{\mathrm{trap}}(10) = 0.00$).

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